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Wireless mechanical and hybrid thrombus fragmentation of *ex vivo* endovascular thrombosis model in the iliac artery



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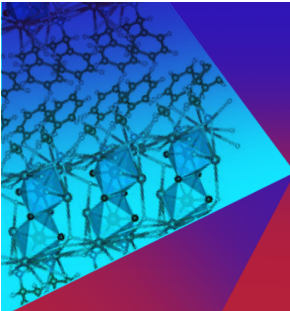
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ABSTRACT

This study investigates the efficacy of an untethered magnetic robot (UMR) for wireless mechanical and hybrid blood clot removal in *ex vivo* tissue environments. By integrating x-ray-guided wireless manipulation with UMRs, we aim to address challenges associated with precise and controlled blood clot intervention. The untethered nature and size of these robots enhance maneuverability and accessibility within complex vascular networks, potentially improving clot removal efficiency. We explore mechanical fragmentation, chemical lysis, and hybrid dissolution techniques that combine mechanical fragmentation with chemical lysis, highlighting their potential for targeted and efficient blood clot removal. Through experimental validation using an *ex vivo* endovascular thrombosis model within the iliac artery of a sheep, we demonstrate direct revascularization of a 13-mm-long, 1-day-old blood clot positioned inside the left common iliac artery. This was achieved by deploying a UMR into the abdominal aorta within 15 min. Additionally, both mechanical fragmentation and hybrid dissolution achieve a greater volume rate of change compared to no intervention (control) and chemical lysis alone. Mechanical fragmentation exhibits clot removal with a median of 0.87 mm³/min and a range of 2.81 mm³/min, while the hybrid approach demonstrates slower but more consistent clot removal, with a median of 0.45 mm³/min and a range of 0.23 mm³/min.

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I. INTRODUCTION

Peripheral arterial diseases such as thrombosis pose a significant challenge in modern healthcare, contributing significantly to the development of severe cardiovascular conditions. This phenomenon,

characterized by the formation of blood clots within blood vessels, poses a substantial risk to global health, contributing to high mortality rates and long-term disability.¹ In the context of vascular diseases, thrombosis plays a central role, leading to conditions such as acute

limb ischemia (ALI).² ALI is a medical emergency that occurs when there is a sudden decrease in blood flow to a limb, threatening its viability. It is typically caused by a blood clot or embolism that blocks an artery or, less frequently, by severe injury or infection. Symptoms of ALI include sudden and intense pain in the affected limb, often felt even when at rest. The limb may appear pale (pallor) and feel colder compared to other parts of the body due to the reduced blood flow. Additionally, the affected limb might show signs of pulselessness, where the pulse is absent or significantly diminished. Patients may also experience paralysis, indicating loss of muscle function, and paresthesia, which includes tingling, numbness, or other abnormal sensations. Immediate treatment is crucial to restore blood flow and minimize tissue damage.³ Treatment options for ALI encompass a spectrum of modalities, spanning from medical therapy to revascularization procedures. These options may involve medications to dissolve the clots, surgical intervention to remove the blockage, or procedures to create a bypass around the blocked artery. Prompt and effective treatment is vital to prevent permanent damage or the loss of the limb. Patients with ALI face a significantly heightened risk of limb amputation, cardiovascular complications, and mortality.

The selection of intervention depends on factors such as the extent of ischemia and concurrent health conditions. Revascularization, whether achieved through surgical procedures or endovascular techniques,⁴ plays a pivotal role in restoring optimal blood circulation to the affected limb. This restoration is crucial for enhancing the prospects of limb preservation. However, in cases where thrombosis occurs in regions that are difficult to reach with tethered flexible systems,⁵ such as small or tortuous vessels, catheter-based thrombolysis and aspiration approaches (e.g., Penumbra) may not be feasible. In such scenarios, traditional methods may also increase the risk of distal embolization, leading to complications like microthrombosis in the lower limb. While distal embolic protection devices and catheter-directed tPA lysis can mitigate these risks, they may not always provide complete protection, emphasizing the need for alternative interventions that can access these challenging regions effectively.

Recent advancements in untethered magnetic robots (UMRs) offer a significant improvement over current mechanical and pharmacological techniques by providing a versatile and minimally invasive approach to navigating and treating vascular conditions in areas that are difficult to reach or inaccessible with traditional tethered systems.^{6–13} Wang *et al.* have proposed a wireless stent-shaped magnetic soft robot for navigating and performing medical interventions in distal vascular regions, facilitating new minimally invasive endovascular procedures.^{14,15} Similarly, Yang *et al.* have introduced a UMR capable of high-speed upstream movement in blood vessels and performing various biomedical tasks, such as drug delivery and tissue biopsy.¹⁶ Jin *et al.* have introduced a catheterization-integrated swarming microrobotic platform for targeted embolization treatment of aneurysms, utilizing pH-responsive self-healing hydrogel microgels guided by ultrasound and fluoroscopy imaging, and triggered by an external magnetic field for effective embolization.¹⁷ Similarly, Pozhitkova *et al.* have introduced a reprogrammable temperature-responsive soft UMR for minimally invasive thrombosis treatment. They efficiently extract plasma clots in vein-mimicking flow conditions.¹⁸ These innovations hold promise for revolutionizing the treatment landscape of ALI and other vascular conditions, offering novel approaches to address complex clinical challenges.

Despite advancements in UMR technology, their behavior within occluded vessels and atherosclerotic plaques remains largely unexplored, especially in *ex vivo* and *in vivo* settings.¹⁹ This gap in research may be due to the complex and dynamic nature of vascular environments, as well as the challenges in accurately simulating and observing UMR interactions within real or *ex vivo* tissue. Additionally, while UMRs are navigated based on 2D imaging techniques such as x-ray fluoroscopy imaging, their operation within the 3D structure of blood vessels presents additional challenges that need further investigation, especially concerning their potential impact on the *tunica intima* layer of the vessel wall.^{20,21} Therefore, as a necessary next step in the development of UMRs, a critical need is to investigate how UMRs can effectively engage with *ex vivo* tissue to advance toward medical management strategies for conditions like ALI.

In this study, we aim to address the limitations of current thrombus treatment methods by exploring a wireless approach using UMRs (Fig. 1). Our goal is to treat thrombi in peripheral arterial regions that are difficult to access with traditional tethered systems, such as microcatheters and microwires, which may struggle to navigate complex vascular networks. The contribution of this study lies in demonstrating the feasibility of using UMRs for targeted thrombus removal within an *ex vivo* endovascular model. We provide a novel comparison between mechanical fragmentation and hybrid approaches, illustrating how the combination of mechanical fragmentation and chemical lysis can enhance clot removal efficiency. This is the first time UMRs have been tested in the iliac artery model with real-time x-ray guidance, showcasing their potential for precise and minimally invasive vascular interventions. Our approach uses x-ray-guided magnetic fields to control the motion of UMRs toward blood clots in an *ex vivo* endovascular thrombosis model within the iliac artery of a sheep. The iliac artery was selected for its size and accessibility, making it an ideal candidate for an initial validation (Sec. VI). Its relatively straight configuration allows for controlled experimentation and precise assessment of clot removal without the added complexity of tortuous vascular networks, which can introduce variability in clot integrity and interaction. By using the iliac artery, we ensure that the integrity of the blood clots is preserved throughout the study, enabling us to focus on the performance of the UMRs and accurately measure their effectiveness in clot removal. This approach lays the groundwork for future studies involving more complex and less accessible vascular regions. First, we

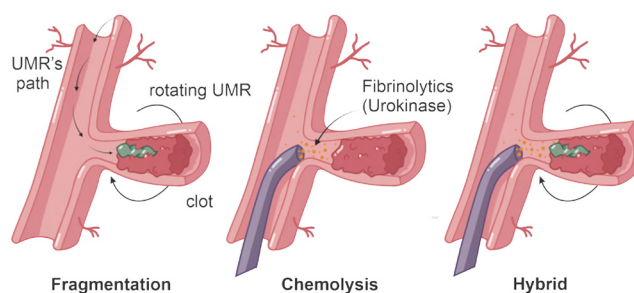


FIG. 1. The engagement with the blood clot for volume reduction and revascularization of blood flow is achieved through three methods: mechanical fragmentation using an untethered magnetic robot (UMR), chemical lysis employing fibrinolytics, and a hybrid approach that combines both fragmentation and chemical lysis to mitigate the risk of peripheral emboli or hemolysis.

demonstrate the capability of biocompatible UMRs to swim controllably against and with blood flow and target blood clots inside the left or right common iliac artery, as shown in Fig. 2(a). Second, we employ cone-beam computed tomography (CBCT) scans to collect anatomical data on size, removal rate, and shape for three distinct groups of blood clot removal approaches. These approaches include the conventional method of thrombolysis through chemical lysis, as well as mechanical fragmentation using UMRs and a hybrid approach. In chemical lysis, clot dissolution is achieved through the action of fibrinolytic agents, which break down fibrin within the clot. Mechanical fragmentation involves physically disrupting the clot structure, which can release trapped blood cells and fibrin fragments, and is likely to contribute to adenosine release due to hemolysis. The hybrid approach combines

both mechanical fragmentation and chemical dissolution, aiming for a synergistic effect to achieve more efficient clot reduction. In this hybrid approach, the fragmented clot is then chemically lysed, leading to complete clot removal rather than just fragmentation. This method minimizes the risk of smaller clots causing peripheral emboli or hemolysis.

The three approaches are tested using an *ex vivo* endovascular thrombosis model (Sec. VI). When deploying a UMR within the iliac artery of this *ex vivo* model, its magnetic moment enables wireless motion control and controlled navigation.²³ The UMR is inserted into the endovascular thrombosis model through a large-bore cannula (19 Fr). X-ray-guided magnetic fields are applied using a robotically controlled rotating permanent magnet (RPM) and the C-arm imaging system, as shown in Fig. 2(a). The endovascular thrombosis model is

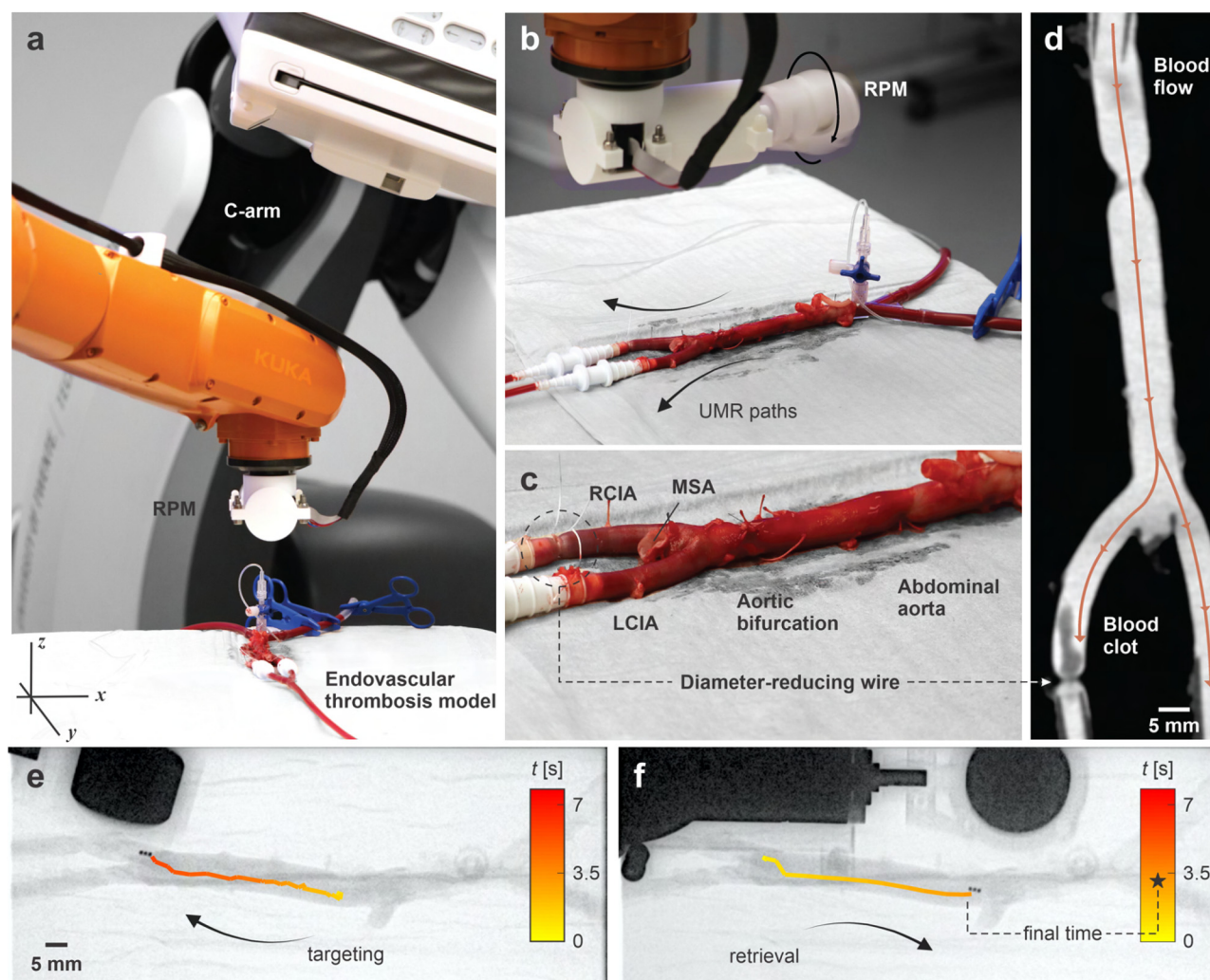


FIG. 2. Examining the impact of mechanical fragmentation and a hybrid approach, which combines fragmentation with chemical lysis, in an *ex vivo* endovascular thrombosis model within the iliac artery. (a) x-ray-guided robotic platform facilitates wireless actuation and localization of untethered magnetic robots (UMRs).²² (b) Rotating magnetic fields are generated to maneuver the UMR toward a clot. (c) The clot is developed and fixed within the right common iliac artery (RCIA), secured in place using a diameter-reducing wire. Blood is allowed to flow from the abdominal aorta through the left common iliac artery (LCIA). The middle sacral artery (MSA) is cut from the model. (d) The size of the blood clot is reconstructed using scans, with the bright region indicating the UMR. (e) and (f) The trajectory of the UMR is tracked using x-ray fluoroscopy images as it travels from the abdominal aorta toward a blood clot inside the RCIA (supplementary material Appendix, Movie S1).

positioned between the detector and emitter of the C-arm, enabling the RPM to apply the desired magnetic torque on the UMR, while blood flows at a predetermined rate [Figs. 2(b) and 2(c)]. For instance, the blood clot is secured within the right common iliac artery (RCIA) using a diameter-reducing wire, enabling blood flow through the left common iliac artery (LCIA). Here, the rotation axis of the RPM is aligned parallel to the aorta and then adjusted before the aortic bifurcation to direct it toward the clot inside the RCIA. The CBCT reconstruction in Fig. 2(d) reveals a 13-mm-long blood clot within the RCIA. One-day-old blood clots, formed from sheep blood, are secured within the RCIA using diameter-reducing wire, and x-ray fluoroscopy images are captured during the direct teleoperation of the UMR, as shown in Figs. 2(e) and 2(f).

II. RESULTS

A. Targeting and direct revascularization

Figure 2(e) illustrates the trajectory of the UMR from the abdominal aorta through the aortic bifurcation toward the blood clot. Along this trajectory, the UMR advances toward the clot with the flow (23 ml/min) at an average swimming speed of 14.3 mm/s, driven by a rotating magnetic field at 15 Hz. Once the UMR's movement is obstructed by the clot, the direction of the RPM is reversed, enabling the UMR to backtrack toward its initial position, as illustrated in Fig. 2(f). Along this trajectory, the UMR swims against the blood flow at an average speed of 29.3 mm/s using an actuation frequency of approximately 20 Hz (supplementary material Appendix, movie S1). With this level of control, it is possible to directly teleoperate the UMR against and along low blood flows, guiding it toward a clot along a predetermined path for targeting and retrieval.

Consider, for example, a blood clot fixed inside the LCIA, as depicted in Fig. 3. In this scenario, the UMR must navigate along a straight path through the abdominal aorta, passing the aortic bifurcation to reach the clot. Once the UMR encounters the clot, the actuation frequency is set to 25.5 Hz, and axial fragmentation occurs between the UMR and the proximal end of the clot. Axial fragmentation is characterized by the collinearity of the long axis of the UMR with the long axis of the cylindrical blood clot. In this trial, axial fragmentation is observed for 615 s [Fig. 3(a)], followed by tapering of the clot. The tapering is indicated by the UMR's motion, which tends to rotate about the clot while spinning along its long axis. The clot tapers for 3 s, likely resulting in a blunt peak formation. Subsequently, the UMR moves between the clot and the internal wall lining of the LCIA, as depicted in Fig. 3(c). In this instance, the UMR applies lateral fragmentation, where the side of the UMR and the clot are in contact, as it advances slowly forward. This behavior is observed for at least 25 s, after which the UMR ultimately moves past the blood clot, as illustrated in Fig. 3(d) (supplementary material Appendix, Movie S2 and Movie S3). While there is a potential risk of vessel wall injury during UMR interactions, optical examination was used as a preliminary assessment in this study because the primary focus is on the UMR's ability to target and engage with blood clots. Although initial optical assessments of the internal wall lining in four *ex vivo* models showed no visible damage from the UMR, this does not fully rule out the possibility of arterial dissection, perforation, or thrombosis. The targeting and engagement with blood clots inside an *ex vivo* endovascular thrombosis model in the iliac artery has not been previously achieved, making this study a significant advancement. However, as a next step,

a comprehensive histological analysis is necessary to thoroughly evaluate any potential damage to the vessel wall.

Figures 3(a) and 3(d) illustrate the initial engagement of the UMR with the blood clot and its subsequent passage beyond it, respectively. This experiment underscores the feasibility of achieving direct revascularization of obstructed blood vessels in 15 min. The tapering action of the UMR likely facilitated its sliding movement past the clot. However, achieving volume reduction in such a short time frame is unlikely. Therefore, the volume of the blood clot is reconstructed using CBCT scans over an extended period of fragmentation time.

B. Mechanical fragmentation dissolution approach

The UMRs used incorporate a magnetic core, enabling their bodies equipped with helical fins to swim or screw through viscous fluids and viscoelastic materials,^{24,25} respectively (Sec. VI). There are two key design features that significantly influence both swimming and mechanical fragmentation performance. (1) Ferromagnetic material: The amount of ferromagnetic material within the UMR is critical because it directly determines its magnetic moment and responsiveness to external magnetic fields, affecting both its torque generation and the ability to maintain controlled navigation. Increasing the amount of ferromagnetic material enhances the magnetic torque, enabling the UMR to generate higher forces, which are necessary for effective mechanical clot fragmentation (Secs. IV and VI). However, increasing magnetic material also increases the UMR's weight. (2) Helical fin configuration: The configuration of the helical fins plays a key role in both swimming and clot interaction. UMRs with a discontinuous helical structure produce less drag torque compared to continuous design. This enables the UMR to maintain better control over navigation, especially in high-flow environments. Furthermore, helical fin geometry, such as the number of fins, their pitch, and the overall shape of the robot's head, directly affects the fragmentation process, with larger surface areas increasing contact with the clot and improving fragmentation action. These design considerations were further optimized based on previous studies, where 18 different UMR designs were tested, evaluating head shapes, fin configurations, and magnetic core volumes on clot disruption performance.²⁶ This research supports our design choices, emphasizing the need to balance UMR size, shape, and magnetic properties for effective clot removal.

Increasing the size of this magnetic core improves the magnetic torque and is likely to enhance the UMR-clot interaction, facilitating axial or lateral fragmentation and sliding through the clot during direct revascularization. However, enlarging the magnetic core increases weight and reduces buoyancy, potentially causing unwanted contact with the internal wall lining. The swimming speed of a UMR with one to three inserted 1 mm³ cubic permanent magnets is illustrated in Figs. 4(a)–4(c). In this experiment, the UMR swims inside a water-filled lumen with a diameter similar to that of the iliac artery, and its swimming speed is determined using camera footage. The asymmetric design of the UMR (Materials and Methods) causes the swimming speed to be influenced not only by the flow direction but also by the orientation of the UMR with respect to this flow. The asymmetry is intentional to ensure efficient fragmentation with the conical tip.⁸ However, irrespective of the fluid flow direction, backward swimming, where the UMR advances with its flat side leading, exhibits slower speeds compared to forward (conical tip leading) swimming across all flow rates and configurations of affixed permanent magnets.

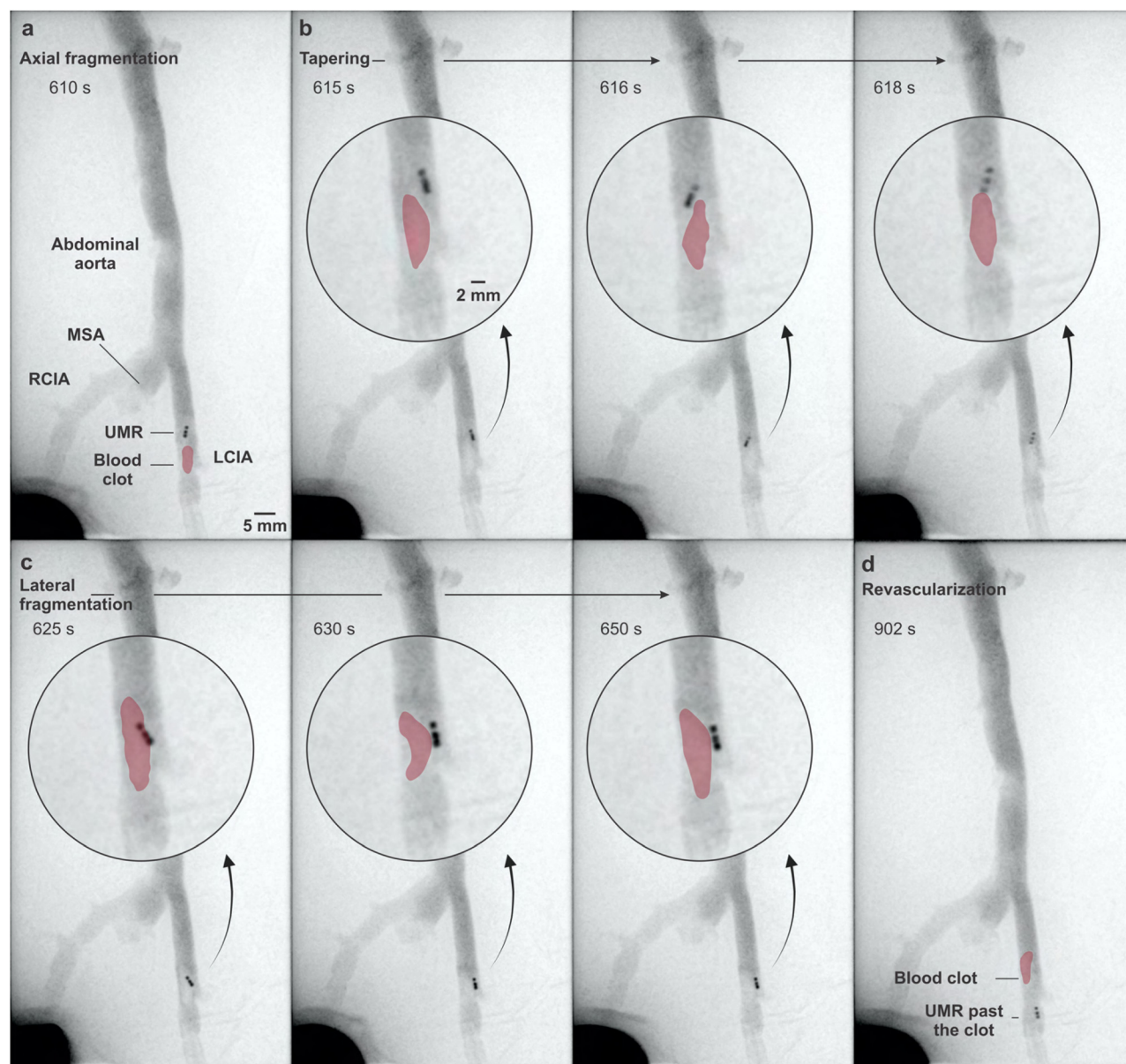


FIG. 3. The untethered magnetic robot (UMR) navigates along a blood clot within an ex vivo endovascular thrombosis model in the iliac artery, completing the passage in 15 min. In this trial, a clot with a length of approximately 13 mm is fixed in the left common iliac artery (LCIA), allowing blood to flow through the right common iliac artery (RCIA), while the median sacral artery (MSA) is cut from the model. (a) The UMR engages in axial fragmentation for approximately 610 s upon reaching the proximal end of the clot. (b) A tapering action is observed for 3 s. (c) The UMR engages in lateral fragmentation for 25 s while confined between the clot and the arterial lining. (d) The UMR moves beyond the blood clot, suggesting potential partial revascularization of the artery ([supplementary material](#) Appendix, Movies S2 and S3). The red shaded regions, manually outlined on each x-ray fluoroscopy image, indicate the shape and position of the blood clot.

Furthermore, these measurements demonstrate that the UMRs can hold position against a maximum water flow of 250 ml/min when swimming in the forward direction.

When swimming backward, the UMR can hold position against a maximum flow of 150 ml/min. Since the number of affixed permanent magnets does not negatively affect the swimming speed against the flow, we utilize the maximum amount of magnetic material

(i.e., three affixed cubic permanent magnets) to increase the magnetic torque during axial or lateral fragmentation. Additionally, the discontinuous helical structure of the UMR, comprising segmented helical sections instead of a continuous helix, enhances fragmentation torque by reducing drag torque, thus improving mechanical clot removal. [Figure 4\(d\)](#) illustrates that the drag torque of the continuous helical structure is slightly greater than that of our segmented helices (Sec. VI), enabling

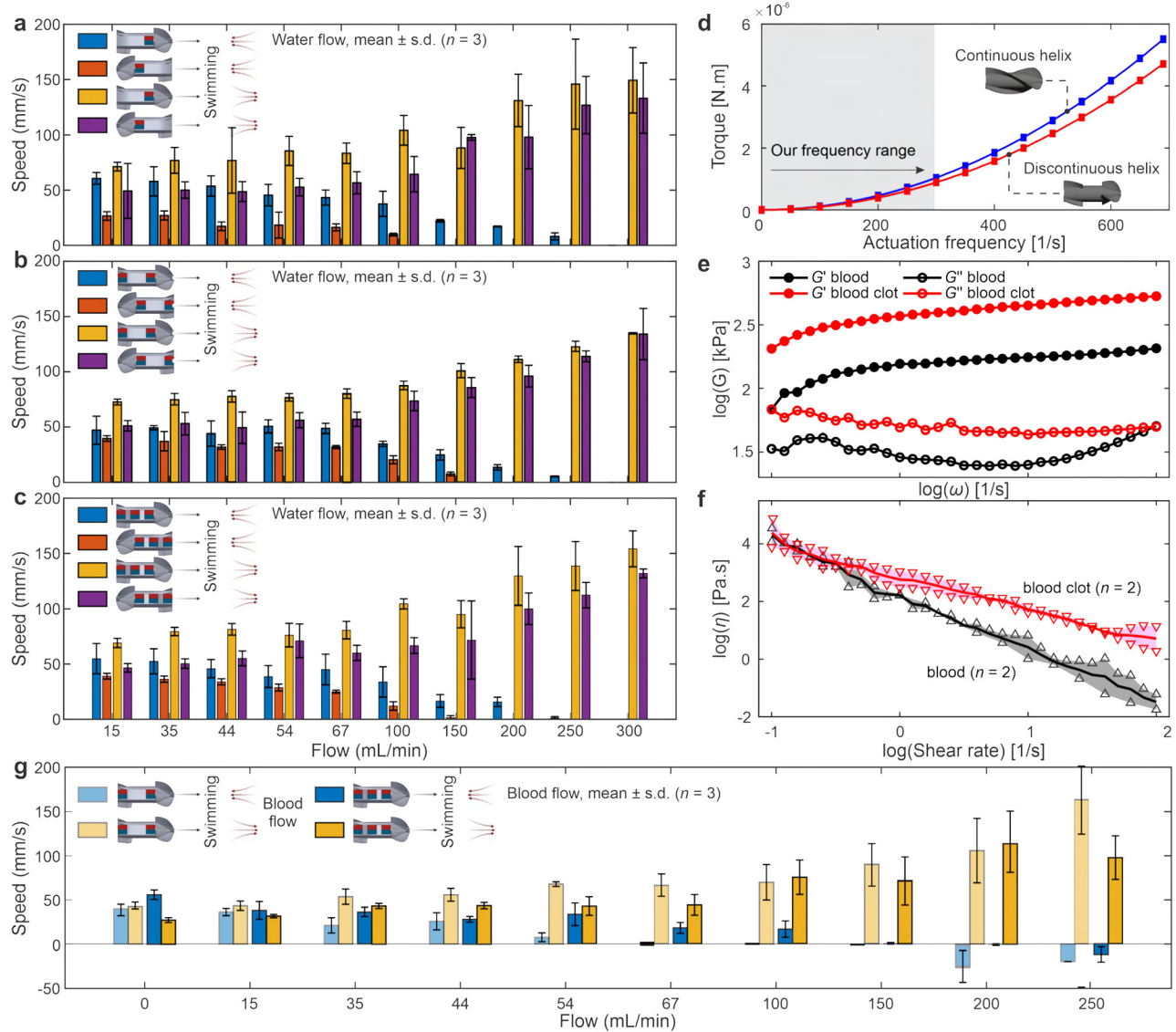


FIG. 4. Flow responses of the asymmetric untethered magnetic robot (UMR) with 1, 2, and 3 magnets in water and blood. (a)–(c) A UMR with single to triple-affixed magnetic cores effectively holds its position against flow rates up to 250 ml/min in water. (d) The discontinuous helical structure of the UMR provides slightly greater torque for fragmentation compared to the continuous helix due to reduced drag torque. (e) Storage modulus G' (filled symbols) and loss modulus G'' (empty symbols) for blood (black symbols) and blood clot (red symbols) at 37 °C. (f) Viscosities η with a shear rate for blood (black) and blood clot (red) at 37 °C. (g) A UMR equipped with two permanent magnets can hold position with blood flow rates of up to 100 ml/min, while one with three permanent magnets can withstand flow rates of up to 150 ml/min.

more torque to be used in mechanical fragmentation. Therefore, we employ an asymmetric UMR design featuring one conical tip for fragmentation, along with three cubic permanent magnets affixed within its discontinuous helical structure, which offers reduced drag torque. Note that blood density is nearly equal to water, so the trade-off between the required magnetic torque and buoyancy is likely to be similar.

Unlike water, blood and blood clots exhibit shear thinning behavior (Sec. VI). Characterization of blood and blood clot deformation and flow properties reveals significant differences between the two [Figs. 4(e) and 4(f)]. Both materials demonstrate predominantly elastic

behavior, with the storage modulus, G' , consistently larger than the loss modulus, G'' , across the tested frequency range [Fig. 4(e)]. Blood exhibits higher viscous properties compared to blood clots, as indicated by the higher ratio G''/G' . The storage modulus increases with frequency, suggesting greater resistance to deformation at higher frequencies. The blood clot displays a significantly higher storage modulus compared to blood, indicating a higher degree of cross-linking in the clot, which tends to increase with the age of the clot. The UMR-clot interactions and rheology are conducted simultaneously on 1-day-old clots to ensure an accurate interpretation of the experimental data.

Viscosity measurements at varying shear rates show a decrease in viscosity with increasing shear rate [Fig. 4(f)], indicating easier flow at higher shear rates. Blood experiences a more pronounced decrease in viscosity compared to blood clots across shear rates, suggesting greater ease of movement for the UMR through blood compared to blood clots. This is confirmed in the experiments, where the UMR progressed at 10 mm/s at 15 Hz actuation frequency with flow in blood and little to no progression in blood clots for an actuation frequency of 20–25.5 Hz with flow ((supplementary material Appendix, Movie S1).

Similarly to the swimming characterization in water, we allow the UMRs to swim in blood with varying flow rates, as shown in Fig. 4(g), and the swimming speed is determined using ultrasound images (Sec. VI). Due to the higher viscosity of blood compared to water, the swimming speed of the UMR is lower across all flow rates. However, when equipped with three affixed cubic magnets, the UMR can hold its position in blood flow rates up to 150 ml/min. Beyond this flow rate, the UMR is dragged along with the blood flow. The UMRs can overcome greater flow rates approaching those encountered *in vivo*²⁷ by increasing the actuation frequency, magnetic field strength, increasing the amount of magnetic material, and decreasing the UMR size (Sec. VI). However, increasing the magnetic field strength or adding more magnetic material to enhance the UMR's capability to overcome higher flow rates also requires an increase in the actuation frequency. In our system, the actuation frequency is limited to 42 Hz. Additionally, the water and blood flow characterization, along with the rheological properties of blood and blood clots, provide valuable insights into adjusting the actuation frequency of the UMR during swimming and upon engagement with the clot. For example, for a given blood flow rate, which can be determined before the deployment of the UMR, we can predict the required time for the UMR to swim from a point of insertion to the target clot.

Figure 5 illustrates the outcome of mechanical fragmentation between a UMR and 1-day-old blood clots fixed into either the RCIA or LCIA. In the case where the blood clot is not influenced by the UMR, its size remains nearly constant over time. Figure 5(a) displays the reconstructed blood clots over 30 min, with CBCT scans obtained every 5 min to estimate volume reduction. The no-intervention (control) group shows an average clot volume reduction rate of $0.26 \text{ mm}^3/\text{min}$ ($\text{SD} = 0.26 \text{ mm}^3/\text{min}$), with a median reduction of $0.31 \text{ mm}^3/\text{min}$ and a range of $0.61 \text{ mm}^3/\text{min}$. This measurement suggests a minimal change in clot size over time without intervention, serving as a baseline for comparison with the results of mechanical fragmentation (supplementary material Appendix, Movie S4). In some of the no-intervention trials, the size of the clot is observed to increase slightly, as shown in Fig. 5(a). This increase is likely due to the clot's ability to trap more blood cells and fragmented clots during blood circulation. The image reconstruction planes (axial, sagittal, and coronal planes) indicate that the volume of the clots remains the same over time. Figure 5(b) shows the measured ratio of the clot volume to the original volume, V/V_0 , over time, indicating a small change in clot volume in the absence of UMR action. The Reye–Archard–Khrushchov wear equation (Sec. VI) is applied to predict the volume rate of change, considering that the volume of the removed debris due to wear is proportional to the work done by friction forces, as shown in Fig. 5(b), providing a reasonable estimation compared to measurements.

When the UMR grinds the clot either axially or laterally, the clot exhibits a substantial average volume reduction of $0.93 \text{ mm}^3/\text{min}$

($\text{SD} = 1.02 \text{ mm}^3/\text{min}$), with a median reduction of $0.87 \text{ mm}^3/\text{min}$ and a range of $2.81 \text{ mm}^3/\text{min}$ (supplementary material Appendix, Movie S5). Figure 5(c) illustrates reconstruction planes of three distinct blood clots from three different *ex vivo* models (different sheep). Unlike the no-intervention trials, significant changes in clot shape are observed over time. The first two blood clots in Fig. 5(c) show a decrease in length, as indicated by the Coronal and Sagittal planes, suggesting that the fragmentation action of the UMR enables volume reduction. However, the third blood clot exhibits less change in volume over time. Despite the variability observed in the results [Fig. 5(d)], likely due to differences in clot composition and the interaction between the UMR and the clot, the mechanical fragmentation approach demonstrated notable efficacy in reducing clot size. The variability of the volume reduction rate during mechanical fragmentation is attributed to the relatively larger deviation in rheological properties of blood clots, as illustrated in Fig. 4(e). This variability in clot composition can be attributed to the use of *ex vivo* tissue and blood clots from four different sheep, contributing to the observed variability in clot reduction rates.

C. Hybrid thrombus dissolution approach

The hybrid thrombus dissolution approach integrates mechanical fragmentation with chemical lysis techniques synergistically to enhance clot removal efficacy. Similar to the no-intervention and mechanical fragmentation experiments, we assess the influence of chemical lysis and hybrid dissolution. For both groups, three different thrombi are assessed. In the chemical lysis group, the fibrinolytic agent (Urokinase) is applied without the UMR, focusing solely on the chemical dissolution of the clot (Sec. VI), as shown in Figs. 6(a) and 6(b). The chemical lysis experiment demonstrates an average removal rate of $0.51 \text{ mm}^3/\text{min}$ ($\text{SD} = 0.56 \text{ mm}^3/\text{min}$) and a median of $0.46 \text{ mm}^3/\text{min}$ and a range of $1.37 \text{ mm}^3/\text{min}$ when Urokinase 100 000 IU Powder is administered and circulated over a 30-minute period, with a new reconstruction of the blood clot every 5 min (supplementary material Appendix, Movie S6). Similarly to the no-intervention trials, the reconstruction planes of the three blood clots in Fig. 6(a) show that the geometry and shape of the clot remain unchanged over time. Conversely, in the hybrid dissolution trials [Fig. 6(c)], the UMR engages with the clot using mechanical fragmentation with the simultaneous engagement of the chemical agent, enabling a combined approach that utilizes both mechanical and chemical mechanisms for clot removal (supplementary material Appendix, Movie S7). This coordinated action ensures that both methods work together synergistically to enhance the efficiency and speed of clot removal compared to using either method alone. With the hybrid approach, the fragmented clot is lysed chemically, leading to complete clot removal rather than merely fragmenting it into smaller clots that could cause peripheral emboli.

The hybrid group exhibits an average removal rate of $0.43 \text{ mm}^3/\text{min}$ ($\text{SD} = 0.10 \text{ mm}^3/\text{min}$, median = $0.45 \text{ mm}^3/\text{min}$, and range = $0.23 \text{ mm}^3/\text{min}$). The higher average removal rate of mechanical fragmentation compared to the hybrid method can be attributed to several factors beyond the simple application of the Reye–Archard–Khrushchov wear law. Fragmentation and wear are influenced by the friction between the surfaces of the UMR and the blood clot. As highlighted in Klaassen *et al.*,²⁸ the frictional behavior is complex; at a constant elastic modulus, friction may increase or

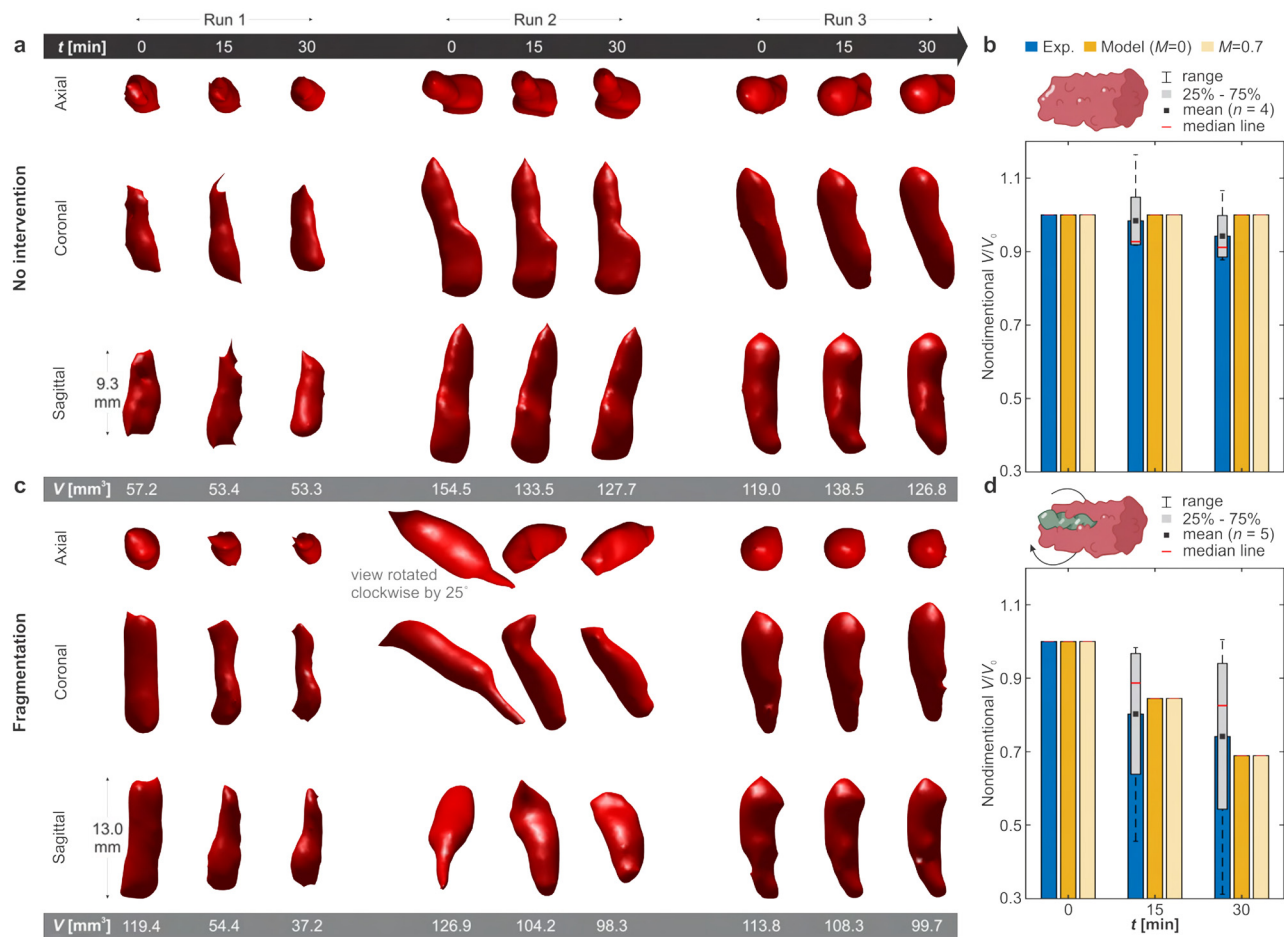


FIG. 5. The UMR-clot interaction is assessed using an *ex vivo* endovascular thrombosis model. One-day-old blood clots are individually inserted beyond the arterial bifurcation, blocking either the left or right common iliac arteries. (a) Volume reduction of blood clots in the no-intervention group over time, with minimal change observed, serving as a baseline for comparison (supplementary material Appendix, Movie S4). (b) The Reye-Archard-Khrushchov wear model predicts zero volume rate of change for the no-intervention case, indicating no significant alteration in clot volume over time. The parameter M is determined based on experimental data. The wear model provides reasonable estimate using $M = 0$ and $M = 0.7$. (c) Volume reduction of blood clots in the mechanical fragmentation group over time, demonstrating reduction attributed to axial and lateral fragmentation interactions between the UMR and the clots (supplementary material Appendix, Movie S5). (d) The large variability in the fragmentation results can be attributed to variations in clot composition, clot size, and the efficiency of UMR interaction with the clot. For $M = 0$ and $M = 0.7$, the Reye-Archard-Khrushchov wear model provides a reasonable estimate of the volume rate of change.

decrease with changes in surface roughness. Similarly, with a constant roughness, friction can be influenced by variations in the elastic modulus. In the context of our study, if the chemical lysis alters the clot's elastic modulus or surface roughness, it could either increase or decrease friction and thereby affect wear and fragmentation. Consequently, further studies are necessary to fully understand the conditions under which hybrid treatment could outperform pure mechanical fragmentation in clot removal. While the average removal rate of the mechanical fragmentation group is higher, the hybrid approach offers more consistent clot removal, with a narrower range of removal rates [Fig. 6(d)]. This suggests that while mechanical fragmentation alone may provide faster clot removal in some cases, the hybrid approach ensures more predictable and consistent outcomes.

The results of our study underscore the efficacy of x-ray-guided UMRs in efficiently removing blood clots within *ex vivo* tissue

environments. Mechanical fragmentation alone demonstrated significant effectiveness, with a median removal rate of 0.87 mm³/min and a range of 2.81 mm³/min. However, the hybrid approach, combining mechanical fragmentation with chemical lysis, yielded more consistent clot removal rates, with a mean of 0.43 mm³/min, a median of 0.45 mm³/min, and a narrower range of 0.23 mm³/min. The observed variability in clot removal rates likely stems from inherent differences in clot composition²⁹ and variations in UMR-clot interactions. Specifically, the proportion of fibrin to red blood cells (RBCs) in the clot can influence its resistance to removal, as fibrin-rich clots tend to be stiffer and more resistant to fragmentation.

Despite these promising findings, it is essential to consider potential complications such as hemolysis, which may occur during UMR procedures. Hemolysis could release adenosine, potentially leading to heart block—a known complication in procedures like Angiojet

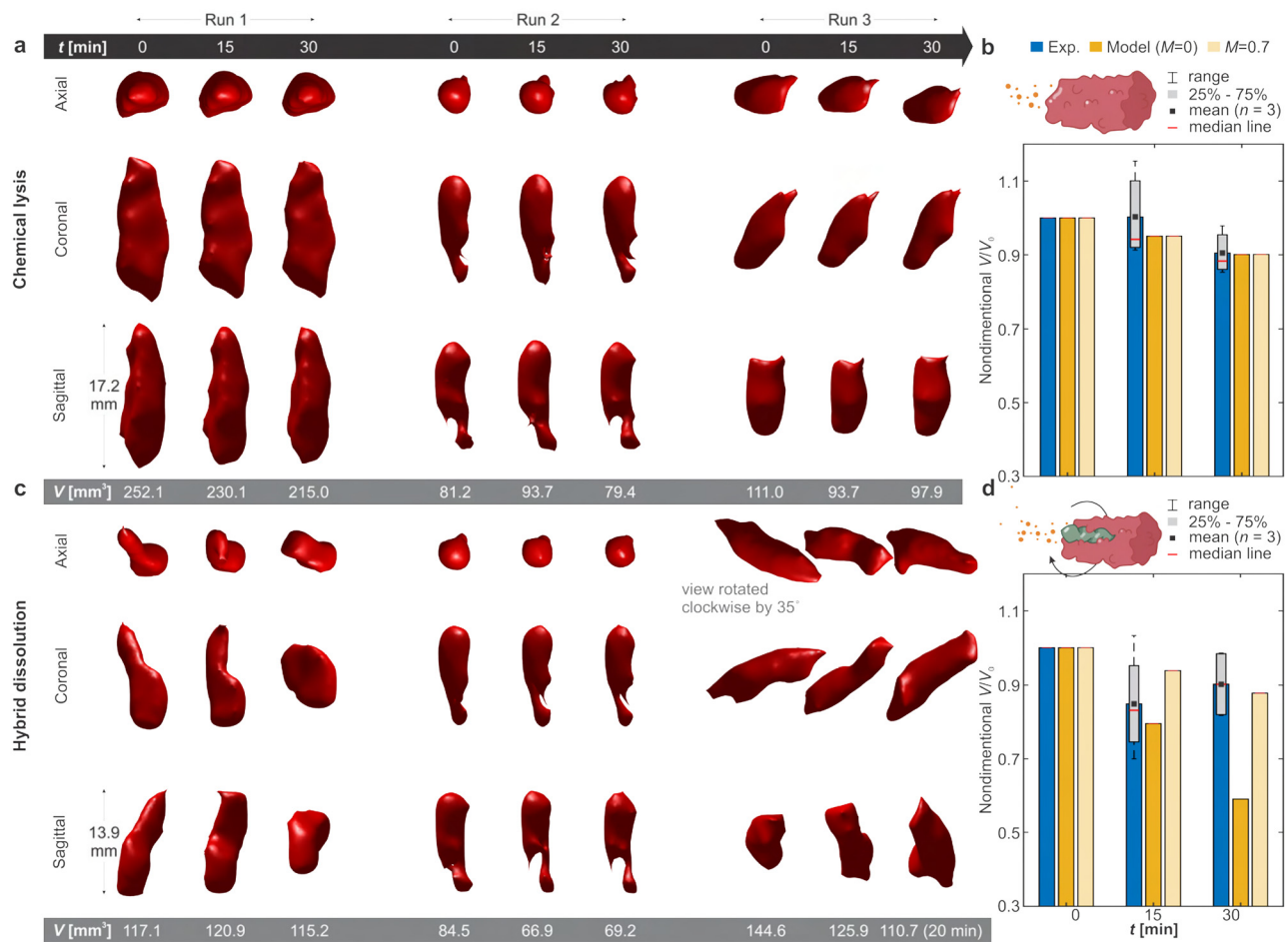


FIG. 6. The UMR-clot interaction is assessed using an ex vivo endovascular thrombolysis model. (a) Volume reduction of blood clots in the chemical lysis group over time, showcasing the impact of chemical dissolution on clot size reduction (supplementary material Appendix, Movie S6). (b) Chemical lysis using Urokinase achieves a volume rate of change of $0.51 \pm 0.56 \text{ mm}^3/\text{min}$. The Reye-Archard-Khrushchov wear model predicts V/V_0 for $M = 0$ and $M = 0.7$. (c) Volume reduction of blood clots in the hybrid group over time, illustrating the combined effect of mechanical fragmentation and chemical lysis on clot removal (supplementary material Appendix, Movie S7). (d) The hybrid approach achieves a volume rate of change of $0.43 \pm 0.1 \text{ mm}^3/\text{min}$. For $M = 0.7$, the Reye-Archard-Khrushchov wear model provides a reasonable estimate of the volume rate of change.

thrombectomy. The degree of hemolysis should be thoroughly explored in future studies to assess the safety of UMRs. This risk is particularly relevant given the observed variability in clot removal rates, which may also affect the extent of hemolysis.

The data in Fig. 7 illustrate clot volume changes over time, allowing for a comparison of the three removal methods. During the initial 30 minutes, a near-zero volume increase in $0.10 \text{ mm}^3/\text{min}$ is observed [Fig. 7(a)]. This period without intervention is followed by 30 min of mechanical fragmentation, resulting in a clot removal rate of $0.88 \text{ mm}^3/\text{min}$. Subsequently, the UMR is retracted for 60 min [Fig. 7(b)] to assess the impact of fibrinolytic drugs on the clot, where we observe a near-zero volume increase rate of $0.15 \text{ mm}^3/\text{min}$. Given the 1-day-old nature of the clots used, the well-developed fibrin network likely contributed to their resistance to thrombolytics. Finally, when the UMR is combined with the fibrinolytic drug, the volume removal rate improves to $0.54 \text{ mm}^3/\text{min}$.

Our study demonstrates the UMR's ability to interact with blood clots and reduce their volume through observed tapering motion, involving rotation around the clot while spinning along its long axis. This dynamic interaction may disrupt clot integrity, potentially releasing trapped blood cells and fibrin fragments. However, we must also consider the risk of clot fragments traveling downstream (embolism), potentially causing occlusion in smaller vessels or distal embolization. This risk can be mitigated using chemical lysis, as demonstrated in the hybrid approach, which promotes the complete dissolution of clot fragments rather than leaving them to travel downstream, making it a safer option. Furthermore, investigation is needed to elucidate the precise mechanism of clot removal and evaluate associated risks comprehensively. Future studies could incorporate high-resolution microscopy and dynamic imaging modalities, to directly visualize UMR-clot interactions and assess the fate of clot material during thrombolysis. However, conducting high-resolution microscopy on ex

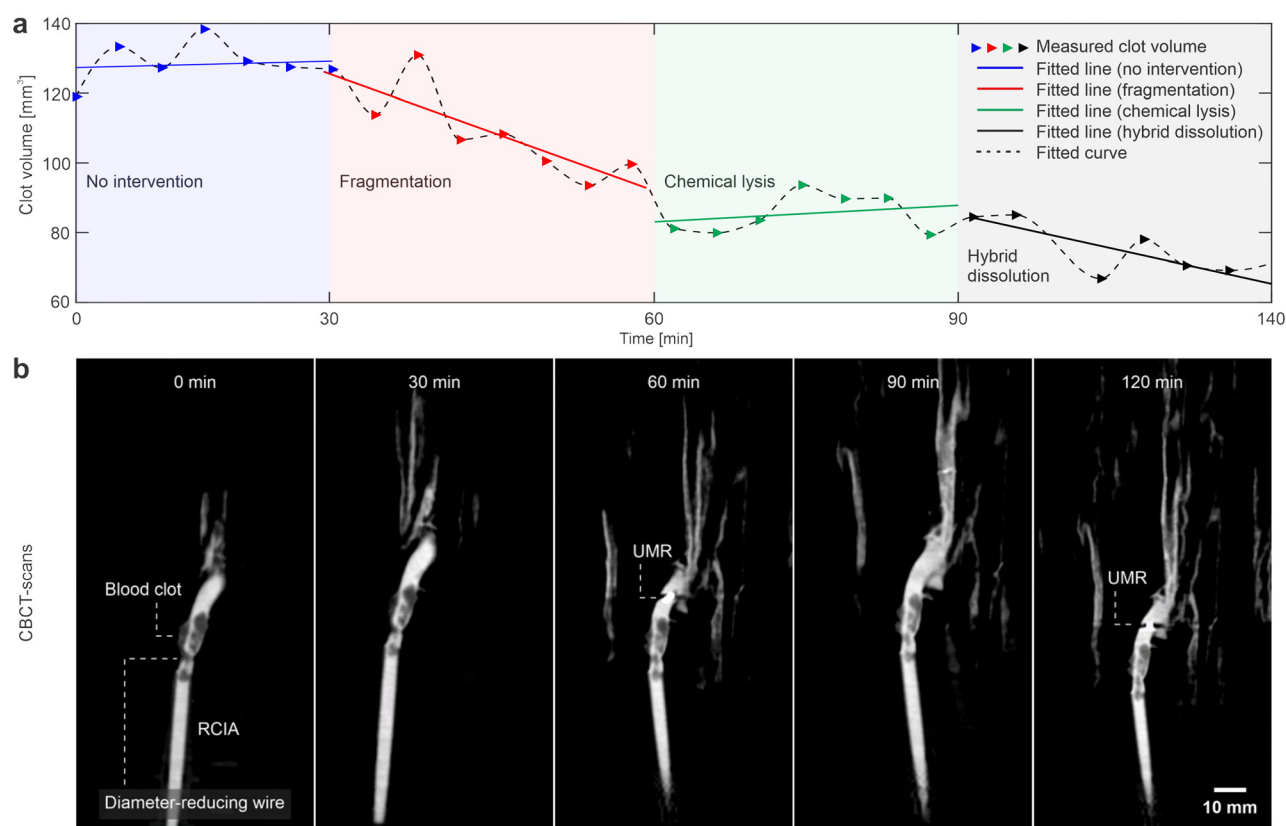


FIG. 7. The volume rate of change is assessed through cone-beam computed tomography (CBCT) scans taken every 5 min for the no intervention, fragmentation, chemical lysis, and hybrid dissolution procedures. (a) Based on the measured construction planes, the volume rate of change is calculated as 0.10, -0.88 , 0.15, and -0.54 mm³/min, for the no intervention, fragmentation, chemical lysis, and hybrid dissolution trials, respectively. (b) The CBCT scans illustrate the location and size of the same blood clot over time.

in vivo models may not be feasible due to limitations in preserving tissue integrity. Additionally, experimental models simulating physiological conditions more accurately, such as *in vivo* animal models, could provide valuable insights into the real-time dynamics of UMR-mediated clot removal and its potential implications for thrombolytic therapy. Furthermore, the mechanical properties of blood clots are likely influenced by hematocrit levels, RBCs concentration, and fibrin content, which in turn affect the clot removal rate regardless of the engagement strategy. To explore this, we developed six types of clots, varying both RBC concentration and fibrin levels.

III. EFFECT OF RBCS AND FIBRIN CONCENTRATION

To examine the rheological properties of thrombi with varying hematocrit and fibrin concentrations, we developed blood clots based on the methodology of Johnson *et al.*,³⁰ described in Sec. VID. During viscosity measurement, the blood clot undergoes significant deformation, as shown in Fig. 8(a), where viscosity decreases with increasing shear rate. The highest viscosity is found for medium hematocrit concentration. Blood with 48% hematocrit concentration has lowest viscosity. At shear rate 1 s^{-1} , viscosity is 34, 330, and 26 Pa s for hematocrit concentrations 17%, 36%, and 48%, respectively. Addition of fibrin has highest effect for 36% hematocrit and decreases

blood clot viscosity. For low hematocrit concentrations, fibrin only increases viscosity at low shear rates. In contrast, for high hematocrit concentrations, fibrin decreases viscosity at low shear rates but increases it at high shear rates. Due to the influence of hematocrit levels on viscosity, the mechanical fragmentation experiments conducted on four animals display variability in the removal rate, as shown in Fig. 5(d).

In oscillatory measurements, which induce only slight deformation in the blood clot, the elastic modulus is highest for clots with a 36% hematocrit level, indicating greater stiffness. In contrast, for clots with 17% RBCs, the elastic modulus is significantly lower than that observed at higher RBC concentrations, as shown in Fig. 8(b). The elastic modulus measurements at a frequency of 1 Hz reveal distinct values across hematocrit concentrations: 77 Pa for 17%, 250 kPa for 36%, and 52 kPa for 48%. No consistent pattern emerges for fibrin's impact on elastic modulus, as it varies depending on frequency and hematocrit levels. Specifically, fibrin appears to reduce elastic modulus at high frequencies for intermediate hematocrit concentrations, increase it at low frequencies for high hematocrit levels, and has negligible impact at low hematocrit levels. The lower elastic modulus at 17% hematocrit may result from isolated blood cells within the matrix, while overlapping RBCs at 38% could rapidly elevate elastic modulus.

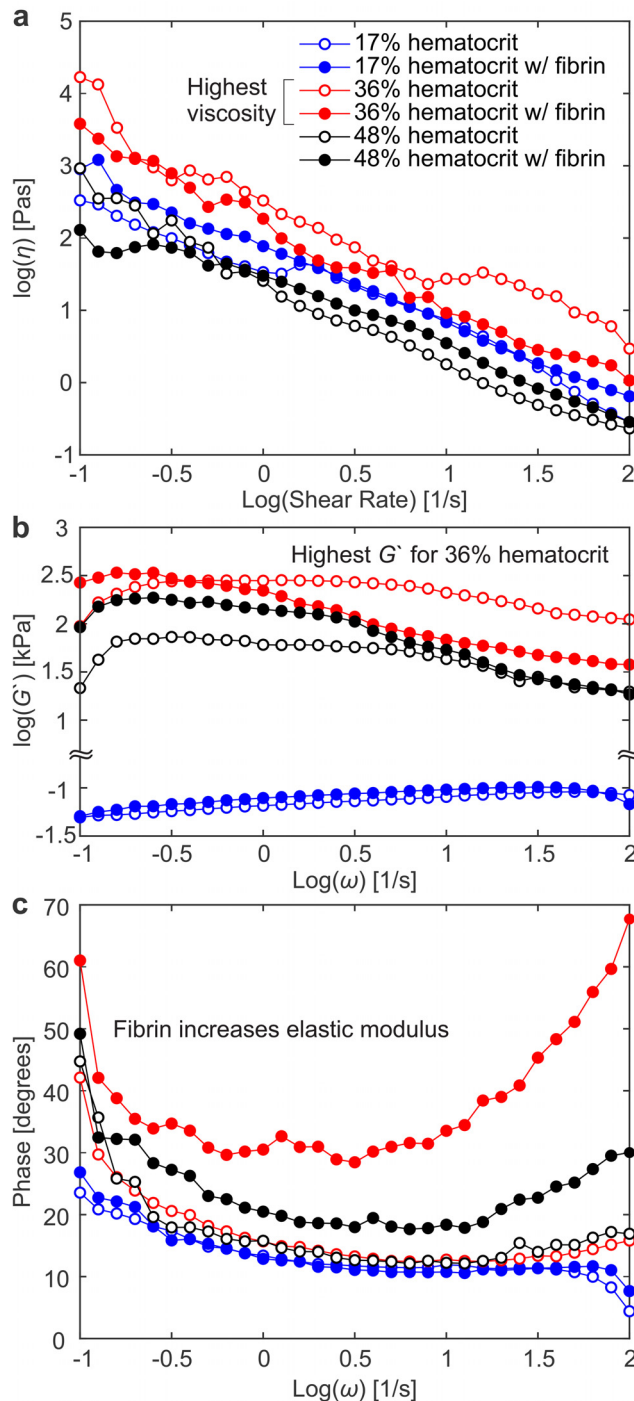


FIG. 8. Rheological characterization of blood clot with different hematocrit concentration and fibrin concentration. Hematocrit concentration is varied from low 17% (blue) to medium 36% (red) up to high 48% (black). Either the clots contain no-added fibrinogen (empty markers) or 0.5 g/l (filled markers). (a) Viscosity vs shear rate for high deformation viscometry measurements. (b) Storage modulus vs angular frequency during oscillatory deformations. (c) Fibrin increases the phase angle of the clot, making it more elastic.

Given the highest elastic modulus for intermediate hematocrit levels, these clots likely exhibit the lowest removal rate. Consequently, our experiments with blood clots of intermediate RBC concentration likely represent the lower bound for removal rate.

Fibrin increases the phase angle of the blood clot, making it more elastic, as shown in Fig. 8(c). This effect is most pronounced at intermediate RBC concentrations, where an optimal ratio of RBCs, surrounding fluid, and fibrin likely allows for enhanced cross-linking within the clot, thereby boosting its elasticity. At low hematocrit levels, however, fibrin's impact on the clot's elastic behavior appears negligible. While the rheological analysis provides valuable insights into the mechanical properties of blood clots and their potential interaction with the UMR, it does not substitute for direct experimental engagement. Testing clots with varying fibrin concentrations and measuring the volume reduction for each group is essential to comprehensively evaluate the efficiency of the UMR under different clot compositions.

To evaluate the influence of fibrin concentration on the removal rate, blood clots are prepared and targeted with UMRs following the same methodology as previous experiments. However, constructing a 3D dissected vessel system for this purpose is impractical, and consistently inducing clots within a non-dissected vessel system presents significant challenges. Instead, blood clots with varying fibrin concentrations are prepared (Sec. VI) and inserted into a cerebral vascular system [Fig. 9(a)]. This approach allows the UMR to demonstrate its ability to target the clot, engage effectively, and return to its point of insertion for retrieval.

UMR-clot interaction measurements were conducted to further investigate the effect of fibrin on blood clot degradation. Each trial consisted of three distinct stages: (1) targeting, where the UMR navigates from the point of insertion toward the clot; (2) engagement, during which the UMR interacts with the clot for either fragmentation or hybrid treatment; and (3) retrieval, where the UMR returns to the point of insertion after completing its engagement with the clot. Clots are inserted and secured in the right ICA by reducing the vessel's diameter, as illustrated in Fig. 9(a). The UMR's trajectory from the point of insertion to the clot is depicted in Fig. 9(b). Between the targeting and retrieval stages, the UMR engages with clots of varying fibrin concentrations to assess its performance under different conditions. These results are visualized in Fig. 10, where the addition of fibrinogen on clots with a hematocrit of 36% is compared for all four groups.

The no-intervention group, as expected, shows no significant decrease in blood clot size, with an average degradation rate of $0.15 \text{ mm}^3/\text{min}$ for clots without added fibrinogen and $0.13 \text{ mm}^3/\text{min}$ for clots with added fibrinogen. In contrast to the no-intervention group, the additional cross-linking within the clot due to the fibrinogen does notably decrease the fragmentation efficiency, and the removal rate drops from $0.88 \text{ mm}^3/\text{min}$ to $0.51 \text{ mm}^3/\text{min}$. This supports the hypothesis that a higher elastic modulus does increase the fragmentation resistance. The addition of Urokinase to the blood as chemical lysis agent did result in higher degradation than the no intervention for both types of clots, yielding a degradation of $1.04 \text{ mm}^3/\text{min}$ for the standard clot and $0.86 \text{ mm}^3/\text{min}$ for the added fibrinogen. This slower degradation for the added fibrinogen can be explained by the mechanism of Urokinase. Urokinase promotes the generation of plasmin, which degrades fibrin network. A bigger fibrin network will take longer to degrade, yielding lower degradation speeds.

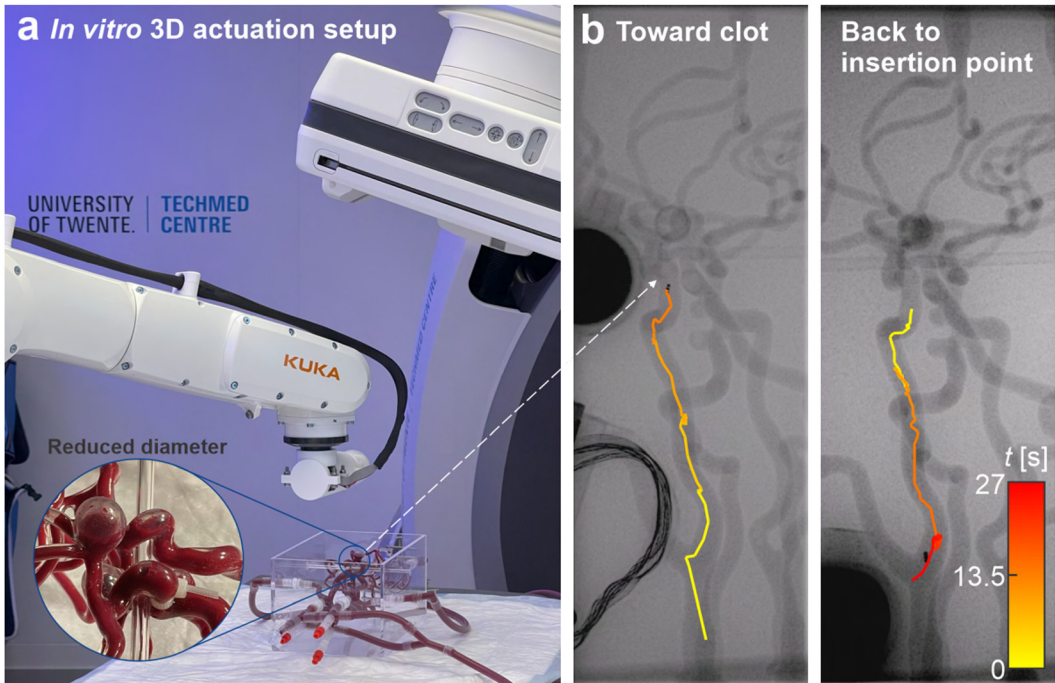


FIG. 9. The clots are trapped at the reduced diameter region within the right internal carotid artery. (a) The imaging platform used to visualize the blood clot inserted within an *in vitro* cerebral vascular network and its surrounding environment. The white dashed arrow indicates the blood clot within the vascular network and in the x-ray fluoroscopy images. The inset shows the blood clot secured in place by the reduced diameter of the vessel. (b) A rotating permanent magnet attached to a robotic arm is used to control the UMR as it swims through the cerebral vascular model toward the blood clot. After successfully engaging with the clot, the UMR is navigated back to its point of insertion.

The hybrid dissolution approach yields most interesting results, where the added fibrinogen results in a significantly higher degradation rate than without the addition of fibrinogen, a difference of $0.93 \text{ mm}^3/\text{min}$ degradation and $0.54 \text{ mm}^3/\text{min}$. The higher efficiency of the hybrid

dissolution approach on the clots with the added fibrinogen can be explained due to the counteractive mechanisms of fibrinogen and Urokinase. Figure 8(c) shows that fibrinogen increases the elastic modulus of the clots, while Urokinase degrades the fibrin resulting in a

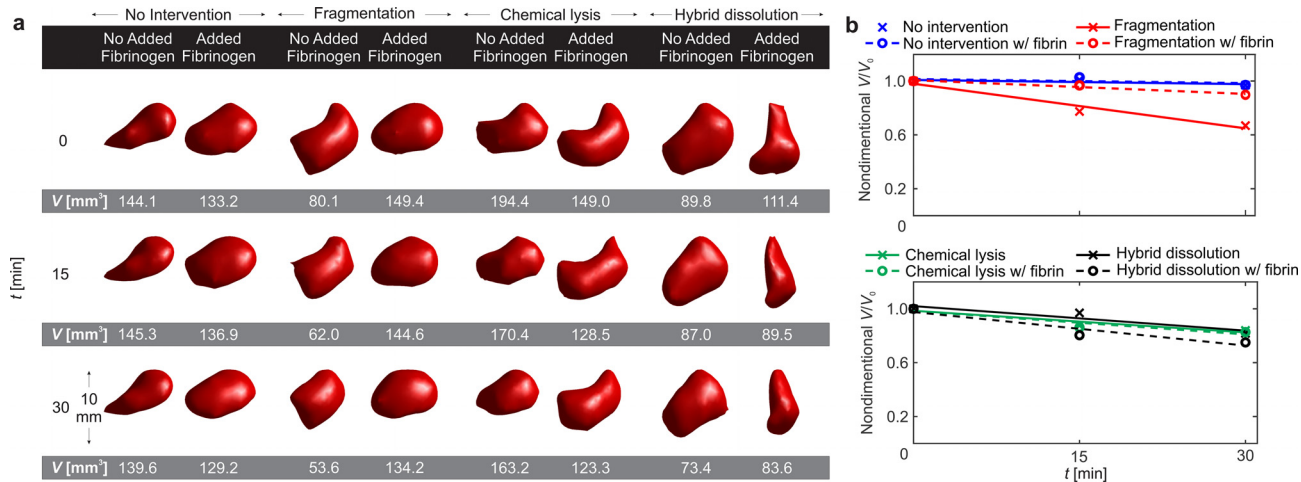


FIG. 10. The UMR–clot interaction and the impact of added fibrinogen are evaluated using an *in vitro* cerebral vascular model. (a) Volume reduction of blood clots across all engagement groups over time, highlighting the effect of added fibrinogen on clot size reduction. (b) The measured nondimensional V/V_0 of the clots over time, the fitted curves show a volume reduction for all but the no-intervention trials, where V/V_0 is the measured ratio of the clot volume to the original volume.

lower elastic modulus. A too stiff clot can be too difficult to fragment, as shown by the difference in degradation efficiency of the fragmentation trials, while a too elastic clot can be too easily deformable to fragment, comparing the no-added fibrinogen trial of the fragmentation approach to the hybrid dissolution approach. There exists an optimum clot stiffness where fragmentation is feasible, in our case the no-added fibrinogen clots, for the fibrin-rich clot, a hybrid dissolution approach is more feasible. These findings align with the results of the *ex vivo* trials, which demonstrate a decrease in efficiency for the hybrid dissolution approach compared to the fragmentation trials. It is highly likely that these clots are more comparable to the no-added fibrinogen clots, as they were similarly prepared without any additional components apart from the blood-preservative adenine-citrate-phosphate-dextrose and its deactivator, calcium chloride.

In summary, hematocrit levels and fibrin concentration influence the mechanical properties of blood clots, impacting both viscosity and elasticity. Clots with intermediate hematocrit levels demonstrate higher viscosity and elastic modulus, which can reduce the consistency of mechanical fragmentation. Fibrin further increases clot elastic modulus, particularly at intermediate hematocrit levels, which enhances clot resistance to purely mechanical methods but makes the hybrid approach more effective by enabling chemical lysis to target the fibrin structure.

IV. DISCUSSIONS

While we acknowledge that the iliac artery model used in this study is less tortuous compared to more complex vascular networks (Fig. 11), it was selected as an essential initial step for validating the UMR's capabilities in an *ex vivo* environment for several key reasons. First, inserting blood clots into an *ex vivo* endovascular system with a straight configuration ensures the preservation of the clot's integrity. Blood clots are viscoelastic [Figs. 4(e) and 4(f)], and navigating them through a tortuous path could alter their structure, thus affecting experimental consistency. By using a more uniform *ex vivo* model, we maintain greater control over the blood clots tested in the no intervention, fragmentation, chemical lysis, and hybrid groups. Additionally, the primary focus of this study is on characterizing the interaction between the UMR and the clot, specifically assessing how the UMR affects clot volume. Accurate measurements of clot volume reduction are critical, and this is best achieved in a controlled, less complex environment. CBCT scans were used to assess these volume changes, and a straight vascular model allowed us to more precisely track and measure the effects of the UMR on the clot over time.

Despite the selection of the iliac artery for this study, the capabilities of UMRs extend well beyond this uniform model. Our experiments in more tortuous networks have shown that UMRs are capable of navigating complex vascular systems. As illustrated in Fig. 11(a), the UMR was able to swim controllably from the common carotid artery

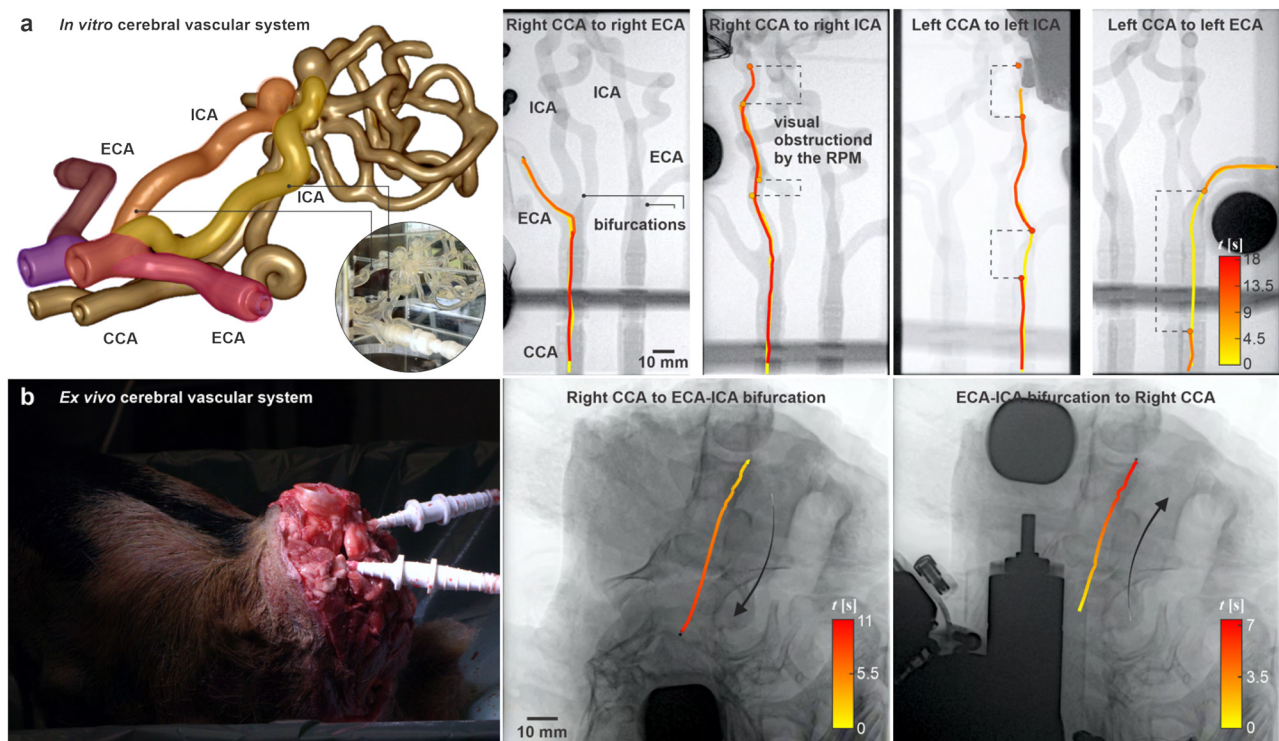


FIG. 11. The UMR swims controllably within the cerebral vascular system, demonstrating its capability to reach and target locations that are typically inaccessible to tethered flexible systems. (a) Within the *in vitro* model, the UMR is deployed inside the right common carotid artery (CCA) and swims toward the distal end of either the right external carotid artery (ECA) or the right internal carotid artery (ICA). Similarly, when deployed inside the left CCA, the UMR navigates within the lumen toward the left ECA and left ICA (supplementary material Appendix, Movie S8). (b) The UMR is deployed inside an *ex vivo* cerebral vascular system of a sheep and is navigated controllably from the arterial bifurcation to the proximal end of the CCA, and vice versa (supplementary material Appendix, Movie S9).

(CCA) to both the right and left external carotid arteries (ECA), and from the CCA to the internal carotid arteries (ICA) within an *in vitro* cerebral vascular phantom (supplementary material Appendix, Movie S8). These results highlight the UMR's ability to navigate within complex, branched blood vessels. Additionally, Fig. 11(b) demonstrates the UMR's successful motion control inside an *ex vivo* cerebral vascular system of a sheep (Sec. VI). In this setup, the UMR moved from the arterial bifurcation (ECA-ICA bifurcation) to the proximal end of the CCA and back, further showcasing its capacity to operate in more anatomically complex environments. This capability demonstrates the UMR's potential to navigate more intricate vascular pathways. If appropriately sized to match the smallest vessel diameter, it may be adaptable for specific endovascular procedures requiring precise access, such as those involving the cerebral vasculature.

These experiments provide compelling evidence of the UMR's potential to address anatomical challenges in regions that are typically inaccessible to traditional tethered catheter systems. Although the current study focused on the iliac artery for its controlled setup, future research will focus on UMR navigation and clot removal in more intricate vascular systems.

V. CONCLUSIONS

In conclusion, our study demonstrates the efficacy of UMRs in decreasing the size of blood clots within *ex vivo* tissue environments. Mechanical fragmentation alone achieved a median removal rate of $0.87 \text{ mm}^3/\text{min}$, representing an average reduction in clot volume of 24.49%, with a range of $2.81 \text{ mm}^3/\text{min}$. The hybrid approach, combining mechanical fragmentation with chemical lysis, yielded a median removal rate of $0.45 \text{ mm}^3/\text{min}$, corresponding to an average reduction of 18.06% in clot volume, with a narrower range of $0.23 \text{ mm}^3/\text{min}$. These results show the potential of UMRs to significantly improve thrombolysis intervention, offering a practical and feasible solution for clot removal. In addition, our study shows the capability of UMRs to achieve direct revascularization within a timeframe of 15 min and successfully navigate past the blood clot.

VI. MATERIALS AND METHODS

A. Experimental design

The experiments utilized an *ex vivo* endovascular thrombolysis model with four sheep (supplementary material Appendix, Table S1). Sheep were chosen for this study because their blood and clots have been shown to be histologically similar to human blood and clots, making them a suitable substitute for coagulation studies. This similarity allows us to draw more relevant conclusions that could be applicable to human conditions.³¹ In each trial, a blood clot was inserted into the LCIA or RCIA, secured with a diameter-reducing wire to facilitate blood circulation. Initial no-intervention (control) tests confirmed clot stability over 30 min. Clot samples were prepared naturally by allowing blood to form clots and were then cut into sections of approximately 15 mm in length. These sections were shaped into cylinders with diameters ranging from 5 to 7 mm to produce samples with consistent volumes and surface areas. The samples could then be further reduced in size to make them fit within the model. Subsequently, the UMR was introduced into the LCIA or the RCIA via a 19 Fr large-bore cannula. An x-ray-guided magnetic field, controlled by a robotically controlled RPM and the C-arm imaging system, was used for navigation. CBCT scans were performed every 5 min to monitor clot size and location

changes over time. The mean, standard deviation, median, and range were calculated for each test group from four different sheep and four biocompatible UMRs.

The iliac artery was chosen as the experimental site due to its straight, accessible structure, which provides a controlled environment for evaluating UMR performance without the additional variables introduced by more tortuous, complex vascular networks. Using the iliac artery ensures that the blood clot's integrity is preserved during insertion, reducing the risk of structural alterations that could occur in a highly curved pathway. This setup allows for precise clot volume measurements via CBCT scans, which are essential for accurately assessing the effects of UMR mechanical fragmentation and hybrid dissolution in a replicable and controlled manner.

B. *Ex vivo* endovascular thrombolysis model in the iliac artery and thrombolytic medication administration

The aorta, renal arteries, and iliac arteries, along with blood, were obtained from euthanized sheep. Four sheep were used in this study. Sheep (*Ovis aries*) were euthanized by injecting 15 ml of Euthasol (AST Pharma, Oudewater, the Netherlands) mixed with 5 ml of Heparin (5000 IU/ml, Leo Pharma, Ballerup, Denmark) into the jugular vein. After confirming death, the sheep were positioned in a Trendelenburg position, and both the jugular vein and the carotid artery were ligated to allow blood to flow into a container. After collecting as much blood as possible, this blood was stored on ice. Subsequently, the sheep were transported to the laboratory, where the vessels were carefully excised and placed in a bag on ice for transport to the simulated operating room, as depicted in Fig. 2. During transport, the blood and excised vessels were kept on ice. Intervention experiments are conducted after one day, allowing us to obtain blood and enable it to clot naturally. These 1-day-old blood clots are inserted either in the LCIA or the RCIA, enabling blood circulation in the model at a controlled flow rate. To prevent the blood clot from moving inside the model, a diameter-reducing wire is wrapped around the LCIA or the RCIA, creating a constriction that prevents the clot from moving from the larger diameter to the smaller diameter of the vessel. The dimensions of the vessels (supplementary material Appendix, Table S1) and the size of the blood clots are extracted from a Cone Beam CT scanner, obtained from four distinct *ex vivo* animal models.

For the chemical lysis trials, Urokinase 100 000 IU Powder, sourced from DeVriMed, Sint Annaparochie, The Netherlands, was used. The medication was reconstituted in 5 ml of water, after which 0.5 ml of the reconstituted solution was added to the blood, totaling approximately 800 ml. This administration protocol aimed to boost thrombolysis and facilitate clot dissolution within the *ex vivo* tissue environment.

To evaluate UMR swimming capabilities in a three-dimensional vascular model, a cerebral endovascular *ex vivo* model is prepared using complete sheep head to closely replicate *in vivo* physiological conditions. This preparation begins with the euthanasia of a sheep via an injection of Euthasol and Heparin into the jugular vein. Once death is confirmed, blood is collected and stored on ice for later use. In contrast to the thrombolysis model in the iliac artery, vessels are left intact, and a continuous two-magnet UMR is introduced from the proximal end of the CCA to navigate the cerebral vascular system directly.

C. Ex vivo characterization of blood and clots

Characterization of blood and blood clot deformation and flow properties were performed through rotational shear-viscosity measurement. For low deformations, storage and loss modulus, G' and G'' , respectively, were measured in the angular frequency range $0.1 - 100 \text{ s}^{-1}$. For both blood and blood clot, the storage modulus is larger than the loss modulus across the entire frequency range. This indicates that both blood and blood clots are predominantly elastic materials. The phase angle is $8^\circ - 26^\circ$ for blood and $5^\circ - 18^\circ$ for blood clot. This indicates that blood has more viscous properties than blood clot. As the frequency increases, the storage modulus rises, indicating greater resistance to deformation at higher frequencies. The storage modulus for blood ranges from 68 to 206 kPa, while for blood clot, it ranges from 205 to 532 kPa. The storage modulus of the blood clot is approximately 2.4–3 times larger than that of blood, indicating a higher degree of cross-linking in the clot compared to blood. The fluid relaxation timescale, τ , is estimated to be 20 s for blood and 30 s for blood clot, based on the results at an angular frequency of 0.1 s^{-1} . Consequently, the Deborah number De is calculated as 2 for blood and 3 for blood clot.

The samples underwent deformation to measure viscosity, η , at shear rates, $\dot{\gamma}$, ranging from 0.1 to 100 s^{-1} . For highly deformable simple materials, the real part of complex viscosity, $\eta'(\omega) = G''/\omega$, is expected to be equal to viscosity $\eta(\dot{\gamma})$ according to Lenz's law. Therefore, η' and $\eta'' = G'/\omega$ are plotted together with η in Fig. 4(d). All curves show a decrease as the shear rate increases, indicating that the materials flow more easily at higher shear rates. The imaginary part of complex viscosity η'' is higher than the real part η' because the samples are predominantly elastic. For blood and blood clots, the viscosity η is much smaller at higher deformation experiments than for smaller deformation experiments, leading to changes in η' and η'' . This indicates that at higher deformations, some bonds in the materials are broken. For blood, the viscosity decreases strongly from $35 \text{ kPa} \cdot \text{s}$ at 0.1 s^{-1} to $60 \text{ mPa} \cdot \text{s}$ at 100 s^{-1} , which is more than five orders of magnitude. The viscosity also decreases strongly for blood clots from $7.7 \text{ kPa} \cdot \text{s}$ at 0.1 s^{-1} to $1.9 \text{ Pa} \cdot \text{s}$ at 100 s^{-1} but less than for blood. At a shear rate of 0.32 s^{-1} , both materials have the same viscosity. These measurements are used in the Reye–Archard–Khrushchov fragmentation law to predict the clot volume rate of change.

D. Preparation and evaluation of blood thrombi in cerebral vascular models

Sheep blood was collected directly into a container preloaded with adenine–citrate–phosphate–dextrose at a 1:9 ratio to prevent premature coagulation. Following collection, the blood was centrifuged at 3000 g for 10 min to separate components. Platelet-poor plasma (PPP) and RBCs were then isolated into separate containers. By adjusting the volumes of PPP and RBCs, blood samples with hematocrit levels of 17%, 36%, and 48% were created. Each sample was placed in a 15-ml centrifuge tube containing 2.06% calcium chloride solution in a 1:9 ratio. To produce samples with different fibrin concentrations, no fibrinogen was added to the first batch, while $0.5 \times 10^{-3} \text{ kg/l}$ of fibrinogen was added to the second batch. After preparation, the clot mixtures were left to coagulate overnight at 37°C .

For the rheological analysis, the clots were cut into pieces approximately 8 mm in length. These clots were then compressed until they filled the entire parallel plate, after which the rheology measurements were started. For the *in vitro* degradation trials, the clots were formed

in 1.8-ml cryogenic tubes and cut into a piece of 10 mm in length and split along its long axis. These clots were then inserted into the cerebral vascular model (Anatomical Vascular Model, HN-S-A-010, ELASTRAT, Switzerland) via the right CCA. After insertion, the right ECA was clamped shut and blood flow was induced to move the clot to its location within the right ICA. A cable tie was put around the right ICA to trap the blood clot within the model, as shown in Fig. 9(a). For the fragmentation and hybrid dissolution groups, a UMR was inserted into the model's right CCA. After insertion, the UMR was navigated toward the clot using the control method described in Sec. VII. Upon reaching the clot, the UMR engaged with it for a duration of 30 min, after which it was navigated back to its initial insertion point, as illustrated in Fig. 9(b). The blood clots were replaced in between each trial.

E. Fluidic and structural effects

The UMR is a rigid body equipped with three helical fins at the head and three at the distal end with a magnetic moment $\mathbf{m}$ oriented perpendicular to its long axis. Various factors, including fluid velocity, vessel walls, and fluid flow within the vessels, influence the velocity of the UMR. When the UMR is subjected to an external magnetic field $\mathbf{B}$, a torque $\mathbf{m} \times \mathbf{B}$ is created. This torque causes the UMR to rotate with an angular rotational speed ω and move forward with a velocity U . The velocity is dependent on the UMR's geometry, rotational speed, and fluid properties such that

$$U \propto R_{\text{cyl}} \omega \varepsilon_{\text{cyl}}^2 f(De, \beta), \quad (1)$$

where R_{cyl} and ε_{cyl} are average UMR radius and amplitude of helical surface protrusions normalized by R_{cyl} . The fluid is characterized by the Deborah number $De = \tau\omega$, where τ represents the fluid relaxation timescale, and $\beta = \eta_s/\eta$ is a ratio of the blood serum viscosity to the total viscosity of blood.^{32,33} The fluid relaxation timescale is estimated from oscillatory shear experiments as

$$\tau = \lim_{\omega \rightarrow 0} \frac{G'}{\omega G''}. \quad (2)$$

The properties of blood and blood clot are measured in a rheometer using parallel plate geometry with a diameter of 25 mm and a gap distance of 1 mm. The temperature is controlled to body temperature, 37°C . Storage and loss moduli are determined by a frequency sweep, which initiates small sample deformations with a shear amplitude of 0.2% and varies the angular frequency from 0.1 to 100 rad/s. Viscosity is obtained from a flow sweep measurement, which varies the flow rate $\dot{\gamma}$ and induces large-scale deformations in the material.

F. Design of untethered magnetic robots

Our UMR was designed using the CAD software Fusion 360. The central part of the UMR is a cylinder with a partial conical tip. The cylinder has a diameter of 1.74 mm. The partial cone tip starts with a diameter equal to the diameter of the cylinder and decreases to 0.51 mm over a length of 1.9 mm. Two sets of three propeller fins are attached to the front and back of the UMR. The fins have a size of 2.03 mm along the length of the UMR and a width of 0.55 mm and thickness 0.15 mm. The external diameter of the UMR (cylindrical body and fins) is 2.84 mm (supplementary material Appendix, Table S2). This size of UMR ensures a good fit with its target, the iliac artery

with an inner diameter of 4.7–9.4 mm. The UMR is small enough to facilitate sufficient control within the iliac arteries while still being in proportion with the blood clot size. We compared this design with a UMR that has a single set of three continuous helices by performing simulations using the open-source CFD software OpenFOAM [see Fig. 4(d)]. A laminar model was used and the rotation of the swimmer was simulated by rotating part of the mesh and using the arbitrary mesh interface feature of the software. Results show that discontinuous helices produce less drag torque than continuous helices. Therefore, we choose to use this design because it leaves more torque available to grind the blood clot. Additionally, the difference in drag torque exerted on continuous and discontinuous helices becomes more noticeable as the actuation frequency increases, giving the discontinuous design an advantage at higher frequencies. To enable the UMR to operate effectively at higher actuation frequencies, its step-out frequency can be increased by increasing the volume of the ferromagnetic core and reducing the UMR's size. This approach improves its tolerance to higher flow rates, as illustrated in Fig. 12.

This simulation evaluates the performance of UMRs with two sizes and three magnetic core volumes to assess how these parameters affect swimming under a 3 mT magnetic field at step-out frequency. The baseline UMR, with a 2.8-mm diameter and 1 mm³ magnet volume, reaches a maximum speed of 0.4 m/s at 128 Hz. For comparison,

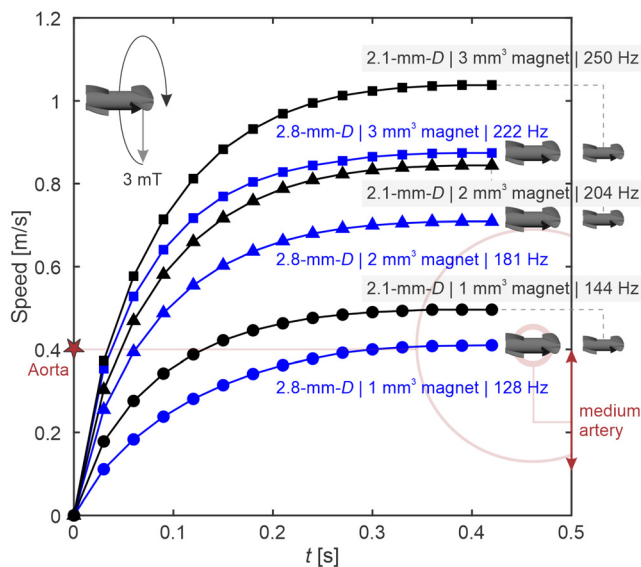


FIG. 12. Performance of two UMRs under a 3 mT magnetic field and step-out frequency conditions is determined numerically. The baseline UMR, with a 2.8 mm diameter and 1 mm³ magnet volume, reaches a swimming speed of 0.4 m/s at 128 Hz. For reference, the sizes of the aorta and a medium artery are shown as red circles, with flow speeds of 0.4 m/s in the aorta (red star) and 0.1–0.4 m/s in the medium artery (red double arrow). A scaled-down UMR at 75% of the baseline diameter, maintaining the same magnet volume, achieves a higher step-out frequency and velocity due to its reduced size. The baseline UMR with double and triple magnet volumes (2 and 3 mm³) also shows an increased step-out frequency and maximum velocity. The highest speed is achieved by the smaller UMR with the largest magnet volume, demonstrating that both increased magnetic volume and reduced swimmer size enhance performance. UMRs with greater magnetic volume or reduced size exhibit swimming speeds that exceed the average flow velocity in both the aorta and medium artery.

a smaller swimmer with 75 % the diameter but the same magnet volume achieves a higher step-out frequency and velocity, owing to its reduced size. Additionally, increasing the magnetic volume to 2 and 3 mm³ in swimmers of baseline size yields higher step-out frequencies and velocities. The results, obtained by numerically solving Newton's second law with Euler's method, indicate that both increasing magnetic volume and reducing swimmer size improve the step-out frequency and maximum speed, with speeds exceeding typical flow velocities in medium arteries and even the aorta.

At the step-out frequency, as the UMR reaches higher velocities, it can counteract a greater flow rate, given by $\mathcal{F} = \pi d_{\text{ves}}^2 v_F / 4$, where v_F is the fluid velocity and d_{ves} is the vessel diameter, assuming plug flow. For a fixed vessel diameter of 8 mm, a UMR with a 1 mm³ magnet core can withstand flow rates of approximately 1.2 l/min for the larger UMR and 1.5 l/min for the smaller UMR. With a higher magnet volume of 3 mm³, critical flow rates increase to 2.6 l/min for the larger UMR and 3.1 l/min for the smaller one. These values are close to the estimated range of human blood flow rates,⁶ typically around 1–1.5 l/min.

Considering their diameters, both larger and smaller UMRs are suitable for navigation within the aorta, medium arteries, vena cava, and veins. Based on simulation results in Fig. 12, the larger UMR with two or three magnets and the smaller UMR can achieve velocities exceeding 0.4 m/s, enabling them to move effectively against flow in all these vessels. Given that flow velocities can reach up to 0.4 m/s in the aorta and medium arteries, the larger UMR with a single magnet would be able to withstand flow in these vessels but may lack the speed needed to advance against it.

G. Modeling of fragmentation, lysis, and hybrid dissolution

The simulation of clot fragmentation by the rotating UMR incorporates the concept that the volume, V , of the removed debris is directly proportional to the work done by friction forces, following the Reye–Archard–Khrushchov fragmentation law:³⁴

$$Q = \frac{dV}{dt} = -\frac{KW\mathcal{L}}{H}, \quad (3)$$

where Q represents the total volume of fragmentation debris produced per time, K is a dimensionless constant, W stands for the total normal load, $\mathcal{L}$ indicates the sliding distance, and H signifies the hardness of the softest contacting surface. The total normal load correlates with the UMR's translational speed, where $W \sim U$. Similarly, the sliding distance is proportional to the frequency of rotation, denoted as $\mathcal{L} \sim f$. The hardness of the blood clot is assumed to be influenced by the elastic modulus, represented by G' , so $H \sim G'$. Therefore, the fragmentation of particles representing the blood clot is described by

$$Q = -\frac{K_g U f}{G'}. \quad (4)$$

Blood clot volume reduction occurs at a constant rate K_l when lysis molecules are in proximity to the blood clot and we have

$$Q = -K_l. \quad (5)$$

The combined model incorporates an interaction parameter M to adjust the interplay between fragmentation and lysis such that

$$Q = -(1 - M) \left(\frac{K_g U f}{G'} + K_l \right). \quad (6)$$

For scenarios where there is no intervention by lysis and fragmentation, the parameters K_l , K_g , and M are all set to zero. Consequently, Q equals zero, indicating no volume reduction. In the case of fragmentation alone, K_l equals zero and M equals zero. The value of K_g is determined to best match experimental data. Similarly, for lysis alone, K_g is set to zero, and the parameter K_l is adjusted to optimize agreement with experimental data. In the hybrid case, the initial volume reduction rate is computed assuming no interaction ($M = 0$). However, since the interaction is observed in the experimental data, the optimal value for M is determined to align with the experimental results. The average normalized experimental volume reduction rates Q are 0.26, 0.93, 0.51, and 0.43 mm³/min for the no interaction, fragmentation, lysis, and hybrid scenarios, respectively. Notably, the rate is lowest when there is no interaction, yielding $Q = 0$ in the model. The parameter K_g is estimated to be approximately 7.1 $\mu\text{N} \cdot \text{s}$, based on experimental conditions where fragmentation was conducted at 25 Hz, $U = 50$ mm/s, and $G' \approx 571$ kPa. The lysis parameter is determined as $K_l = 0.51$ mm³/min. For the hybrid action of fragmentation and lysis without interaction ($M = 0$), the model predicts the highest volume reduction rate as 1.44 mm³/min, which does not align with experimental findings. Thus, an interaction parameter of $M = 0.7$ is established. A value of M greater than zero suggests that the combination of fragmentation and lysis is less effective than their individual effects.

H. Fabrication and hemobiocompatibility

The UMRs were produced through Masked Stereolithography Apparatus (MSLA) employing a 3D printer (Phrozen Sonic Mini 4K) using Phrozen Aqua-Gray 4K resin. During printing, a single layer height of 50 μm was utilized. Following printing, the components were cleaned with isopropyl alcohol in an ultrasonic bath for 7 min. Post-curing was carried out using an Elegoo Mercury plus curing station for 12 min. UMRs, with a length of 6 mm and a diameter of 2.84 mm, were constructed by assembling a 3D-printed body with one to three enclosed permanent magnets made of NdFe Grade-N45 material (S-01-01-N Supermagnete, Gottmadingen, Germany). Each permanent magnet $1 \times 1 \times 1$ mm³, possessing a magnetic moment of 1.07×10^{-3} A·m². The magnets are positioned such that their magnetic moment is perpendicular to the long axis of the body. This configuration enables the UMR to follow an external weak-strength magnetic field with a maximum field strength of 1.4 mT allowing the UMR to swim within blood upon rotation. This field strength is measured at an RPM-UMR gap of 150 mm. After assembly, UMRs were coated with LipoCoat 4AC coating technology (LipoCoat BV, The Netherlands). Previously, we showed that cell adhesion, cell morphology, focal adhesion formation, cell proliferation, and cell differentiation potential remain unaffected by the coating components.³⁵ We also investigated the biocompatibility of the coating *in vitro*²² in terms of protein fouling and biofilm formation, which both showed $\geq 95\%$ fouling reduction. Various hemocompatibility assays were carried out in-house and by HaemoScan BV, The Netherlands. Hematology testing in whole blood was carried out by HaemoScan BV with standard tests comprising platelet, and red blood cell counts as well as quantification of hemoglobin and material-induced hemolysis. Thrombus tests

included visual and gravimetric assessment of thrombi formed on the samples, quantification via immunostaining of fibrin adsorbed to the samples, and enzymatic quantification of attached platelets. Platelet activation was tested by quantifying released thromboxane B2 and beta thromboglobulin as well as platelet aggregation. Coagulation tests consisted of the quantification of thrombin-antithrombin III complex and fibrinopeptide A. Inflammation and complement activation were assessed by quantifying complement component fragments C3a-desArg and C5b-9 as well as elastase. The coated samples passed all hemocompatibility tests at levels comparable if not better than the negative control.

I. Wireless manipulation and swimming behavior analysis

Wireless actuation is achieved through a robotically controlled RPM-actuator. The mechanism generates a rotating magnetic field utilizing a cylinder permanent magnet (NdFe Grade-N45), measuring 35 mm in diameter and 20 mm in height, featuring a magnetic moment of 20.67 A · m². The rotational velocity of the permanent magnet is managed using a Maxon 18 V brushless DC motor, while its orientation is regulated via a 6-DOF manipulator (KUKA KR-10 1100-2, KUKA, Augsburg, Germany). This wireless manipulation is conducted within a C-arm fluoroscopic room, ensuring that the operator is not subject to radiation exposure during the *ex vivo* trials. From the control room of the C-arm, the operator remotely controls the movement of the RPM.

Characterization of the swimming speed of the UMRs in water is conducted inside a 9.5 mm ID tube. The RPM connected to the 6-DOF manipulator is placed 100 mm, for 1 mm³ magnetic volume, or 150 mm, for 2 and 3 mm³ magnetic volume, above the tube's center preventing vertical motion by the magnetic attraction of the RPM. The RPM's rotation frequency is set to 40 Hz, which is slightly below the limit of the system, and a flow is induced using a peristaltic pump (Masterflex® Ismatec® MCP Standard Digital Peristaltic Pump Drive with PRO-380 pump head, Antylia Scientific, Glattbrugg, Switzerland). The pump is connected to the tube using two 3D-printed filters to prevent the UMR from traversing into the pump. For each flow rate, the movements both with and against the flow were recorded for both orientations of the UMR ($n \geq 3$).

In the *in vitro* navigation experiments, Fig. 11, a continuous two-magnet UMR is controlled in a three-dimensional environment, and a cerebral vascular phantom (Anatomical Vascular Model, HN-S-A-010, ELASTRAT, Switzerland) is used to simulate the complex geometry of cerebral vessels. CBCT scans are employed to capture the vessel structure in detail, providing data (centerline of the vascular route) for the control system to navigate the UMR accurately within the vascular network. For this navigation, magnetic forces controlled by the RPM-UMR gap are utilized to achieve upward motion. By controlling the yaw angle of the RPM the UMR is steered to swim parallel with the centerline of the vessels toward its target.

J. Fluoroscopy and cone-beam CT images

During teleoperation trials, x-ray fluoroscopy images are captured using the Siemens Healthineers Artis Pheno (Erlangen, Germany). The image acquisition occurs at a frame rate of 5 Hz, with an x-ray voltage peak of 55.9 kV and a tube current of 119 mA. Images are

converted into videos and tracked using Tracker 6.2.0³⁶ to visualize the traversed path and the speed of the UMR. For volume analysis, CBCT scans are obtained using the same visualization platform. To enhance the visibility of blood clots within the blood, contrast agent iomeron 300 is added (50 ml/l) to the blood prior. After an initial CT-reconstruction at the start of a trial, CBCT scans are obtained every 5 min for 30 min for each trial. For the mechanical fragmentation and the hybrid dissolution approach, this requires moving the RPM-actuator out of the CBCT's field of view to prevent unclear footage due to x-ray scattering. The CBCT scan data are imported into 3D Slicer V 5.6.1³⁷ to segment the clot and derive its volume. An initial segmentation is made using an intensity range in the Segment Editor that accurately encompasses the blood clot. While the vessel walls often fall (partly) within this intensity range, manual intervention was used to ensure accurate clot segmentation. The Segment Statistics tool is used to compute the volume of the segmented clot. The clot reconstructions are made using Matlab (MathWorks, Inc., Natick, MA, USA), the surface of the clot is colored red, and a light source is added to create shadows to help visualize the three-dimensional structure of the clot.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for details: Table S1. Dimensions of the *ex vivo* endovascular thrombosis model in the iliac artery. Table S2. Dimension of the untethered magnetic robot. Movie S1. The UMR travels from the abdominal aorta toward a blood clot inside the right common iliac artery. Movie S2. The UMR engages with the blood clot through axial fragmentation. Movie S3. The UMR moves beyond the blood clot. Movie S4. Volume reduction of blood clots in the no-intervention group over time. Movie S5. Volume reduction of blood clots in the mechanical fragmentation group over time. Movie S6. Volume reduction of blood clots in the chemical lysis group over time. Movie S7. Volume reduction of blood clots in the hybrid dissolution group over time. Movie S8. UMR navigates within an *in vitro* cerebral vascular system. Movie S9. UMR navigates within an *ex vivo* cerebral vascular system of a sheep.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Marcus C. J. de Boer and Leendert-Jan W. Ligtenberg contributed equally to this work and share first authorship.

Marcus C. J. de Boer: Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (equal). **Leendert-Jan W. Ligtenberg:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal). **Iris Mulder:** Formal analysis (equal); Investigation (equal); Software (equal). **Constantinos Goulas:** Formal analysis (equal); Funding acquisition (equal); Resources (equal); Software (equal). **Anke Klingner:** Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Roger Lomme:** Formal analysis (equal); Investigation (equal); Resources (equal); Validation (equal). **Emily A. M. Klein Rot:** Data curation (equal); Resources (equal); Software (equal); Validation (equal). **Dorothee Wasserberg:** Data curation (equal); Formal analysis (equal); Investigation (equal); Resources (equal); Software (equal); Writing – original draft (equal). **Yitong Lu:** Data curation (equal); Formal analysis (equal); Resources (equal); Software (equal). **Remco Liefers:** Data curation (equal); Formal analysis (equal); Resources (equal); Software (equal). **Joep K. van der Mijle Meijer:** Investigation (equal); Methodology (equal); Software (equal). **Gabriëlle J. M. Tuijthof:** Data curation (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal). **Doron Ben Ami:** Conceptualization (equal); Funding acquisition (equal); Resources (equal); Validation (equal). **Udi Sadeh:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Validation (equal). **Oded Shoseyov:** Conceptualization (equal); Funding acquisition (equal); Resources (equal). **Julein Leclerc:** Data curation (equal); Formal analysis (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal). **Aaron T. Becker:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Pascal Jonkheijm:** Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Resources (equal); Validation (equal). **Michiel Warlé:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal). **Islam S. M. Khalil:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon request.

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